

AD7190

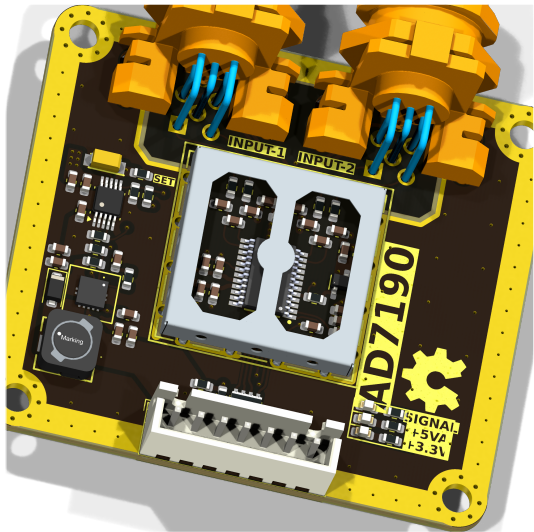
Variant:

2026-09-29

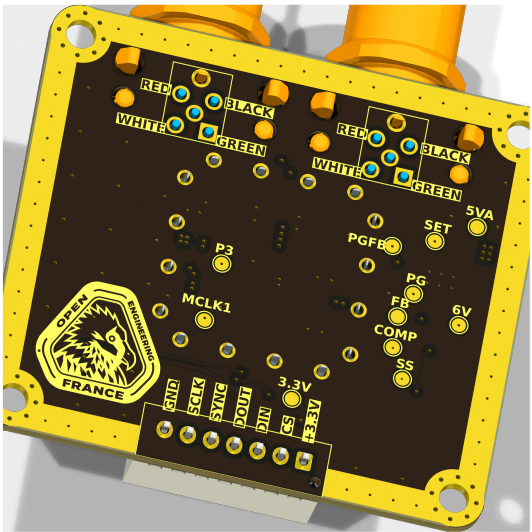
Rev + (Unreleased)

Page	Index	Page	Index	Page	Index	Page	Index
1	Cover Page	11		21		31	
2	Block Diagram	12		22		32	
3	Project Architecture	13		23		33	
4	AD7190_Weight_Sensor	14		24		34	
5	LT3042_6V_to_5VA	15		25		35	
6	TPS61086_3.3V_to_6V	16		26		36	
7	Holes, Fiducials	17		27		37	
8	Connectors	18		28		38	
9	Power - Sequencing	19		29		39	
10	Revision History	20		30		40	

TOP VIEW



BOTTOM VIEW



NOTES

OPEN_WEIGHT - AD7190 24-bit load-cell front-end, part of OPEN THRUST LAB.
Single +3.3 V host supply, SPI interface, 2 x 4-wire bridge inputs.

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 29-Sep-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:

+3.3V host -> TPS61086
6 V -> LT3042 +5VA (AVDD + excitation)

DEBUG NOTE:

TP on every rail.
Wait >= 600 ms (PG) before ADC init.

DESIGN NOTE:

SPI mode 3,
SCLK <= 5 MHz,
RDY on DOUT.

DESIGN NOTE:

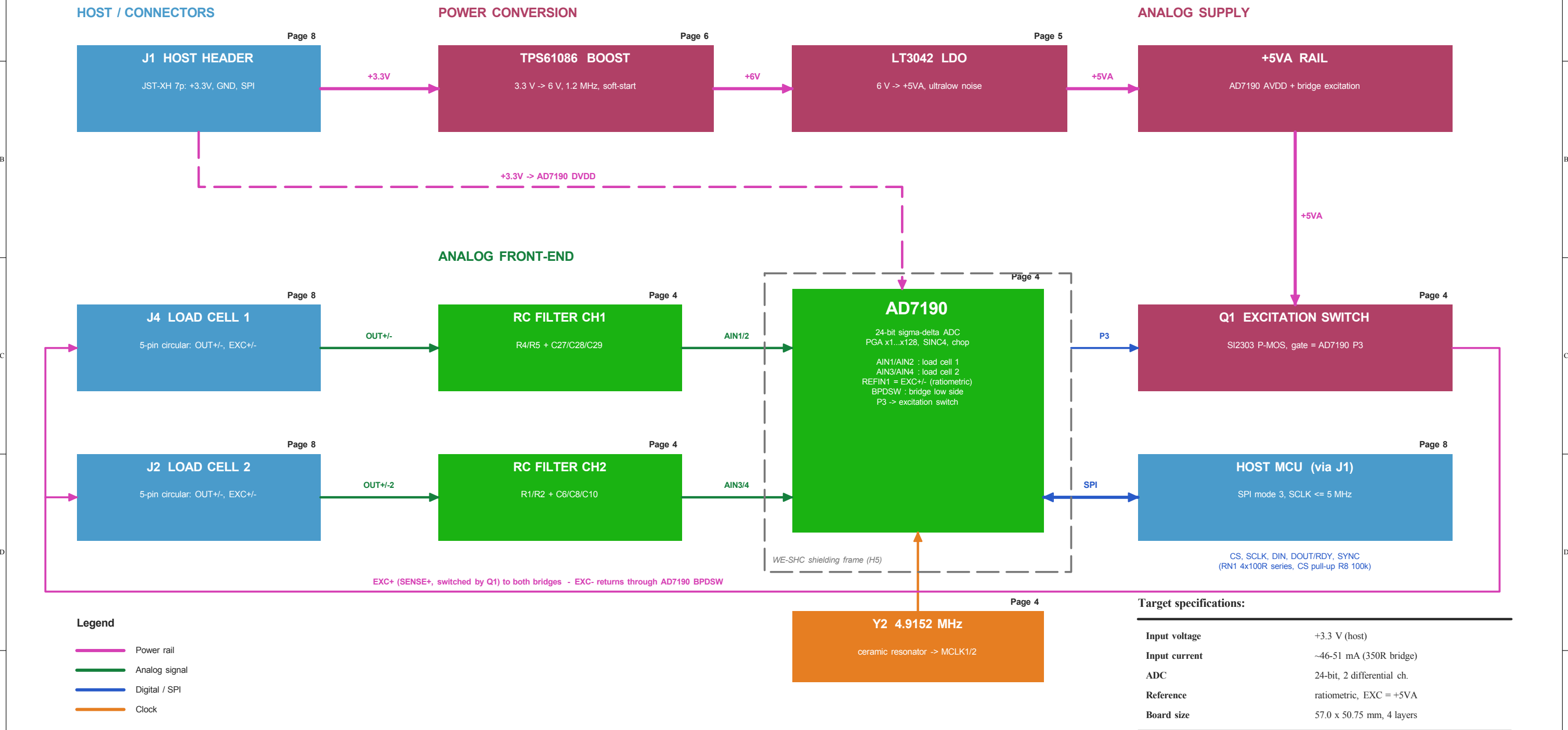
REFIN1 = bridge excitation:
ratiometric.

LAYOUT NOTE:

AFE under the WE-SHC shield H5,
boost SW node away.

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Title: AD7190	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	File Name: OPEN_WEIGHT.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)	
Sheet Path: /		Reviewer:	Size: A3	Sheet: 1 of 10	

[2] Block Diagram

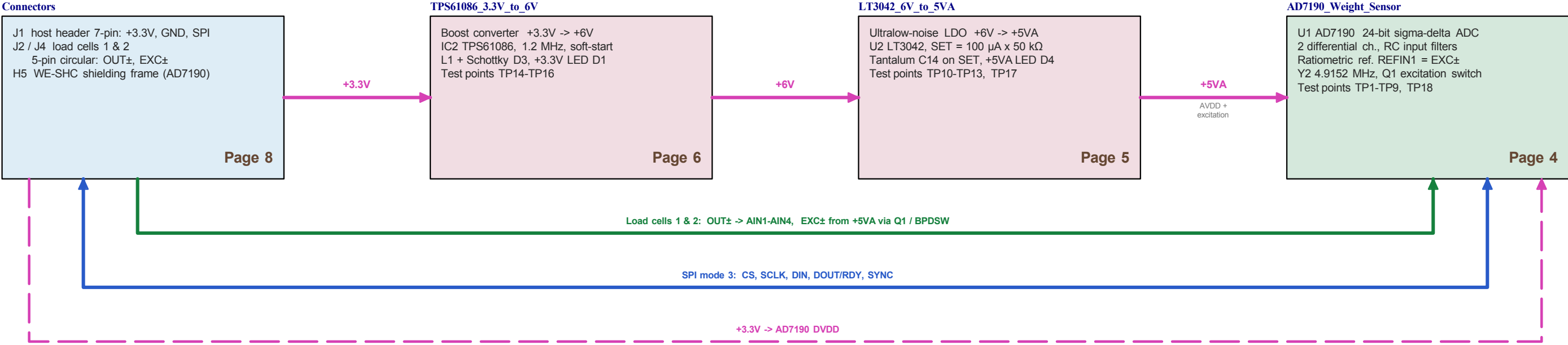


	Comments:		Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Sheet Title: Block Diagram		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /Block Diagram/		File Name: Block Diagram.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
			Reviewer:		Size: A3	Sheet: 2 of 10

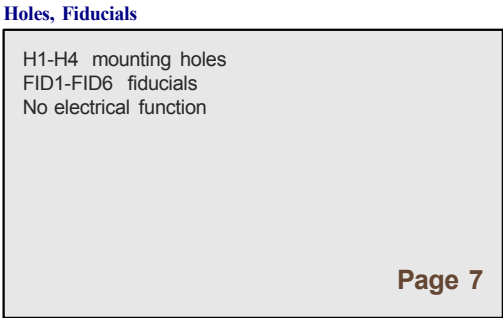
[3] Project Architecture

Sheet hierarchy and signal flow. Click a block or its page number to open the sheet.

HOST INTERFACE



MECHANICAL



LEGEND

- Power rail
- Secondary supply
- Analog signal
- Digital / SPI

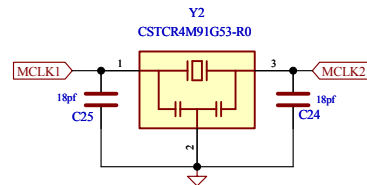
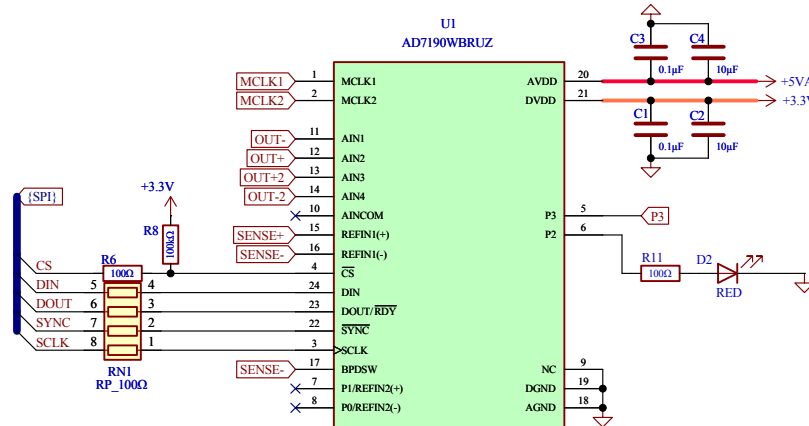
DOCUMENT INDEX

- [1] Title Page
- [2] Block Diagram
- [3] Project Architecture
- [4] AD7190 Weight Sensor
- [5] LT3042 LDO 6V -> 5VA
- [6] TPS61086 Boost 3V3 -> 6V
- [7] Holes, Fiducials
- [8] Connectors
- [9] Power Sequencing
- [10] Revision History



Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
Sheet Title: Project Architecture	File Name: Project Architecture.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
Sheet Path: /Project Architecture/		Reviewer:	Size: A3	Sheet: 3 of 10

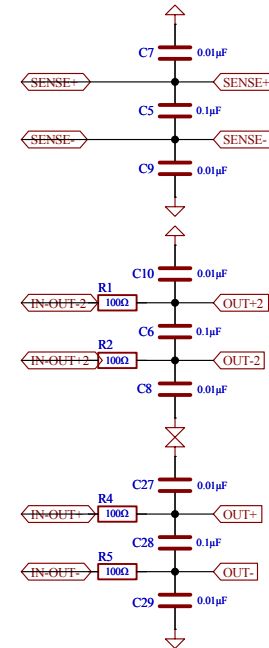
[4] AD7190 - Weight Sensor



**ADC FOR
weighing sensor**

DESIGN NOTE:

Ref Condo for quartz
Murata GRM1885C1H220JA01D
(22 pF, 50 V, C0G, 0603, ±5%)



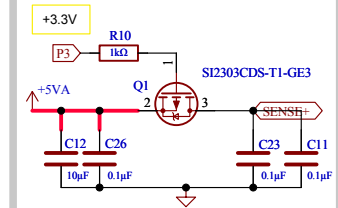
Calcul de filtre : $f_c = 1 / (2\pi \times R \times C) = X \text{ Hz}$

DESIGN NOTE:

Ref Condo for Filtre
Murata GRM1885C1H103JA01D (10 nF, 50 V, C0G, 0603, ±5%)
KEMET C0603R104K5RACTU (100 nF, 50 V, X7R, 0603, ±10%)

Ref Resistor for Filtre
TNPW0603100RBEEA (100 Ω, 0.1 W, thin film, 0603, ±0.1%, 25 ppm/°C)

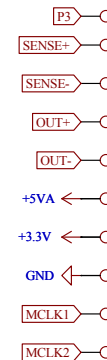
FILTRE



DESIGN NOTE:

IC1 is a P-MOS high-side switch used to enable/disable the load cell excitation. must disabled during calibration

DRIVE GATE



TEST POINT

About

The AD7190 is a low noise, complete analog front end for high precision measurement applications. It contains a low noise, 24-bit sigma-delta ($\Sigma\Delta$) analog-to-digital converter (ADC). The on-chip low noise gain stage means that signals of small amplitude can be interfaced directly to the ADC. The device can be configured to have two differential inputs or four pseudo differential inputs. The on-chip channel sequencer allows several channels to be enabled, and the AD7190 sequentially converts on each enabled channel. This simplifies communication with the part. The on-chip 4.92 MHz clock can be used as the clock source to the ADC or, alternatively, an external clock or crystal can be used. The output data rate from the part can be varied from 4.7 Hz to 4.8 kHz. The device has two digital filter options. The choice of filter affects the rms noise/noise-free resolution at the programmed output data rate, the settling time, and the 50 Hz/60 Hz rejection.



Comments:

AD7190
CSTCR4M91G53-R0
AD7190_TEST_BOARD CN

Company:

OPEN THRUST LAB

Board Name:

AD7190

Sheet Title:

AD7190 - Weight Sensor

Sheet Path:

/Project Architecture/AD7190_Weight_Sensor/

File Name:

AD7190 - Weight Sensor.kicad_sch

Designer:

SIEBERT DIMITRY

Reviewer:

Variant:

Git Hash:

b3867f9

Project Name:

OPEN_WEIGHT

Date:

2026-09-29

Revision:

+(Unreleased)

Size:

A4

Sheet:

4 of 10

DESIGN NOTE:

$V_{OUT} \approx I_{SET} \times R_{SET}$
 $I_{SET} \approx 100 \mu A$

Donc :
 $R_{SET} = V_{OUT} / I_{SET}$
 $R_{SET} \approx 5.0 V / 100 \mu A = 50 k\Omega$
 (=> valeur standard recommandée : 49.9 kΩ)

A

B

C

D

E

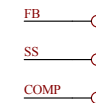
F

F

F

F

F



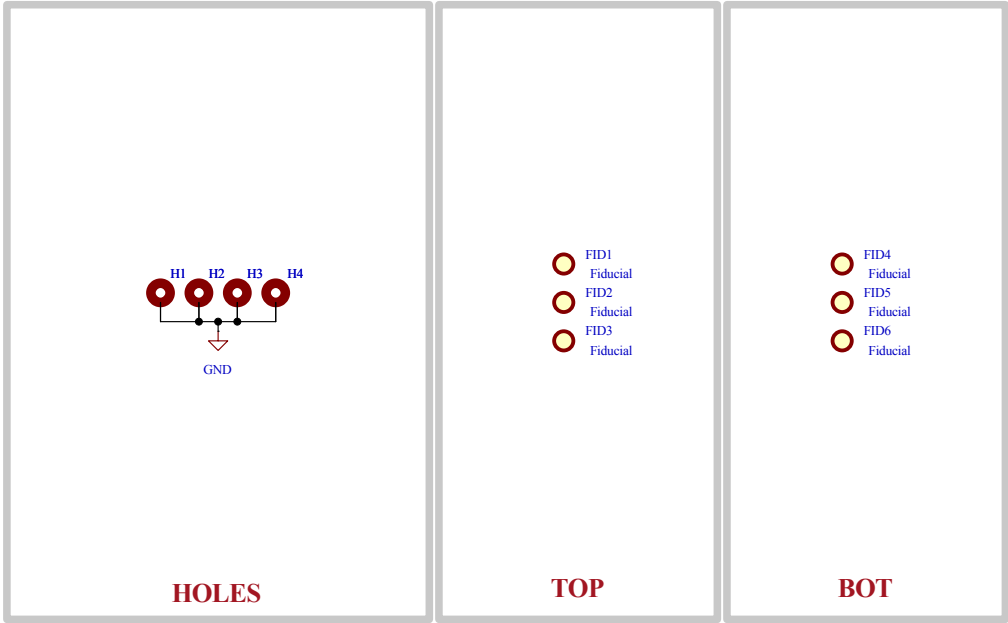
Test points



Pin usage and layout example

of 10

[7] Holes, Fiducials



	Comments: References: 4 x M3 (3.2 mm) plated holes to GND FID1-FID6 not fitted on this revision	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Sheet Title: Holes, Fiducials	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Path: /Project Architecture/Holes, Fiducials/	File Name: Holes - Fiducials.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
			Reviewer:	Size: A3	Sheet: 7 of 10

[8] Connectors

DESIGN NOTE:
J1 host header (JST-XH, 7 pins):
1 +3.3V 2 CS 3 DIN 4 DOUT/RDY
5 SYNC 6 SCLK 7 GND

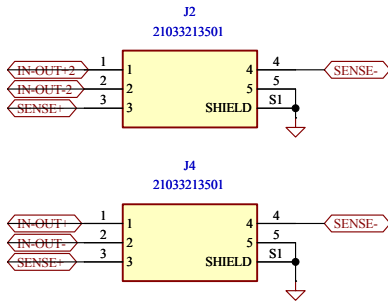
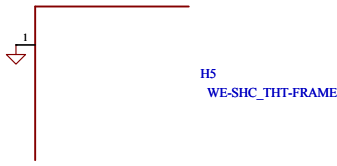
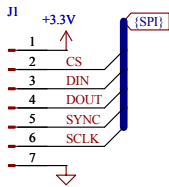
DESIGN NOTE:
J2/J4 load cell (5 pins):
1 OUT, 2 OUT (signal pair), 3 EXC+ (SENSE+),
4 EXC- (SENSE-/BPDSW), 5 GND/shield.
Both channels read V(pin2) - V(pin1):
fix the sign in firmware if needed.

LAYOUT NOTE:
H5 shielding frame covers the AD7190
front-end; connector shells tied to GND.

Pinout and notes



Power flags: +3.3V and GND come from the host connector J1,
(+3.3V and GND come from the host connector J1).



Host and load-cell connectors



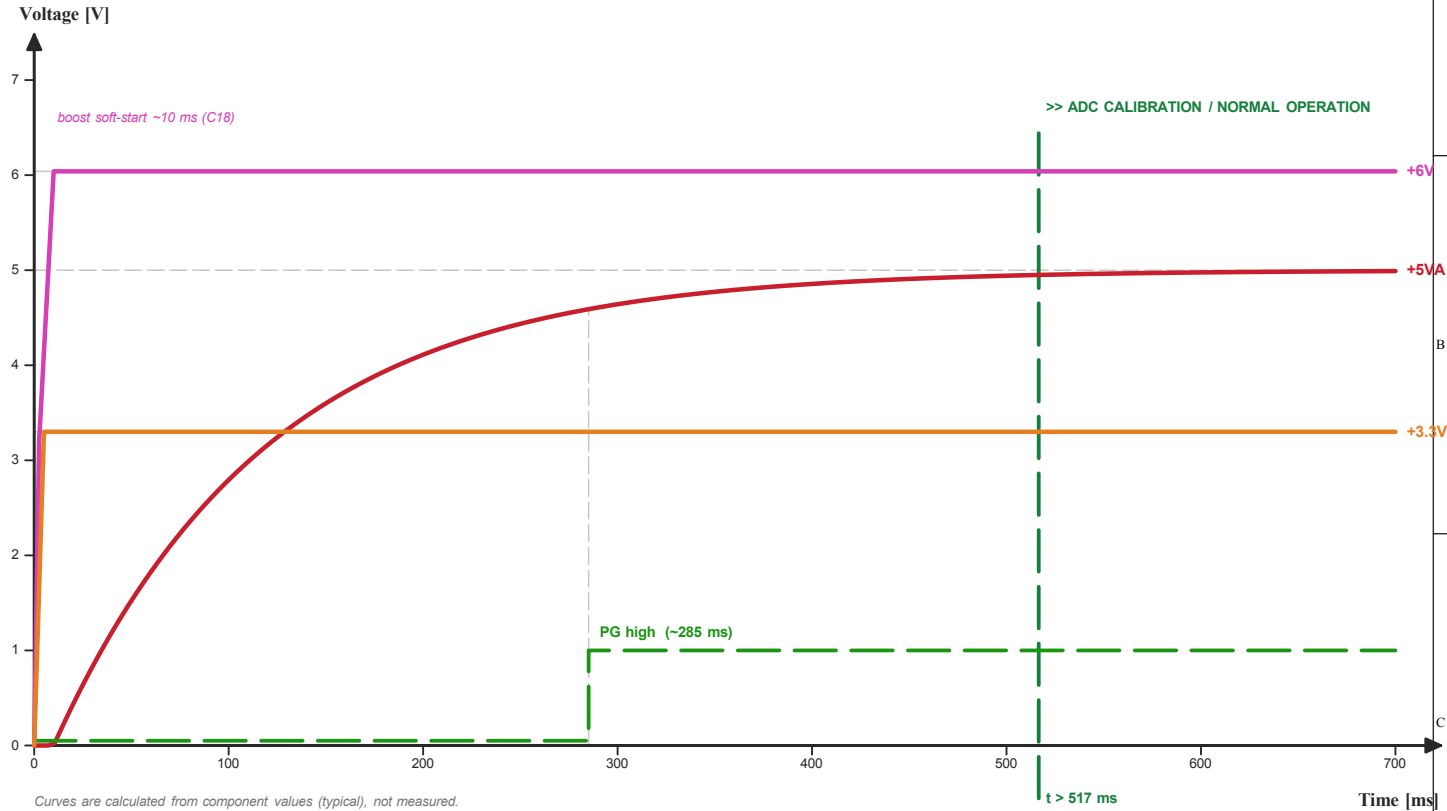
Comments: References: JST XH B7B-XH-AM (J1) J2/J4 21033213501, H5 WE-SHC	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
	Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Connectors	File Name: Connectors.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29
Sheet Path: /Project Architecture/Connectors/		Reviewer:	Size: A3	Revision: + (Unreleased)
			Sheet: 8	of 10

[9] Power - Sequencing

NAME	SOURCE	LEVEL
+3.3V	Host (J1)	3.3 V (host)
+6V	TPS61086 IC2	6.04 V
+5VA	LT3042 U2	5.00 V
PG	LT3042 U2	> 4.59 V

Values from the schematic:
+6V = 1.238 V x (1 + R17/R15) = 1.238 x (1 + 69.8k/18k)
+5VA = ISET x RSET = 100 uA x 50k (R9)
PG threshold = 0.3 V x (1 + R12/R13) = 4.59 V
LT3042 soft-start: tau = RSET x CSET = 50k x 2.2 uF
= 110 ms -> +5VA at 99 % after ~550 ms

DESIGN NOTE:
The LT3042 SET capacitor (C14 = 2.2 uF) gives a 110 ms time constant. Firmware must wait >= 600 ms after power-up (or until PG, TP17) before the AD7190 reset and internal calibration. To start faster, reduce C14 (e.g. 0.47 uF -> tau = 23.5 ms) at the cost of slightly more noise.



	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Power - Sequencing	File Name: Power - Sequencing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 9 of 10

[10] Revision History

Version Unreleased

Added

- AD7190 front-end, 2 load-cell inputs
- +3.3V -> TPS61086 6 V -> LT3042 +5VA
- Self-contained libs and 3D models (lib/)
- JLCPCB 4-layer rules and stackup
- LTspice simulations (Simulation/LTspice)

Changed

- Renamed to OPEN_WEIGHT, KiCad 10

Fixed

- Connectors not linked to ADC in schematic
- AD7190 solder-mask bridges

	Comments:	Company: OPEN THRUST LAB		Variant:	Git Hash: b3867f9
		Board Name: AD7190		Project Name: OPEN_WEIGHT	
	Sheet Title: Revision History	File Name: Revision_History.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-09-29	Revision: + (Unreleased)
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 10 of 10